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TECHNICAL CATALOG · CORRUGATED STEEL PIPE

Metal culvert

Corrugated, galvanized steel structures MiniMultiplate SP MP-68 and Multiplate SP MP-152 / MP-152 S for drainage and road works.

ASTM A929 | LOADS HS20-44 · HL-93 | GALVANIZED

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01 — DESCRIPTION

Corrugated steel pipe (TMC)

They are corrugated, galvanized steel elements, also known as corrugated steel pipe (TMC), made of rings that are assembled on site with high-strength bolts and nuts.

Assembly is simple and needs no specialized labor. Their high structural strength lets them carry deep fills and they work well as surface-water drainage: ideal for carrying water across road infrastructure.

MP-68

MiniMultiplate SP · 68 × 13 mm corrugation · spans up to 1.80 m

MP-152 / 152 S

Multiplate SP · 152 × 51 mm corrugation · spans up to 16 m



SITE Metal culvert installed on site.

02 — MINIMULTIPLATE SP MP-68

Composition and properties

Material requirements per ASTM A929

Chemical composition

The chemical composition requirements of the TMC MiniMultiplate SP MP-68 culvert follow ASTM A929.

TABLE 1 · CHEMICAL COMPOSITION

Element	Composition, %
Sulfur, max.	0.05
Sum of carbon, manganese, phosphorus, sulfur and silicon, max.	0.70

Mechanical properties

The base material has mechanical properties that comply with ASTM A929.

TABLE 2 · MECHANICAL PROPERTIES

Yield strength, min. (MPa)	Tensile strength, min. (MPa)	Elongation in 50 mm, %
230	310	20

Dimensions

Made in two shapes, circular and arch, in thicknesses from 1.80 to 3.50 mm and diameters from 24 to 72 inches. The nominal inside diameters and thicknesses used are:

TABLE 3 · DIAMETERS AND THICKNESSES

Diameter (in)	Diameter (m)	Thickness (mm)
24"	0.6	1.8
36"	0.9	2.0
48"	1.2	2.5
60"	1.5	3.0
72"	1.8	3.3 · 3.5

03 — MINIMULTIPLATE SP MP-68

Corrugation dimensions

Nominal corrugation 68 × 13 mm

The plate corrugations must form smooth, continuous curves. A corrugation is defined by its pitch (crest-to-crest distance) and depth. As shown in Figure 1, the nominal size is 68 × 13 mm.

CORRUGATION REQUIREMENTS

NOMINAL SIZE **68 × 13 mm**	PITCH TOLERANCE **+4.00 / −1.00 mm**
DEPTH TOLERANCE **± 1.00 mm**	NOMINAL INSIDE RADIUS **17 mm**
MINIMUM INSIDE RADIUS **12 mm**	

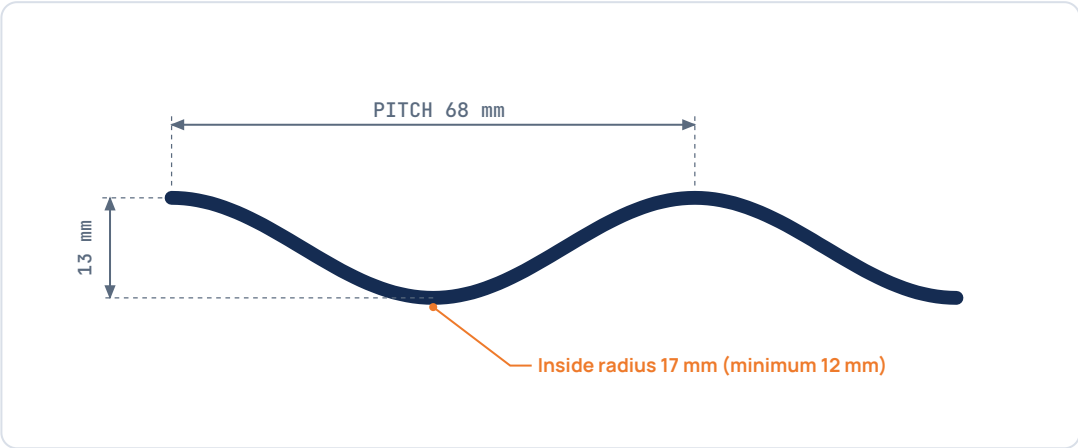


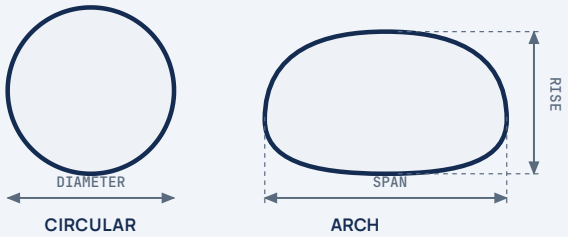
FIGURE 1 Corrugation dimensions 68 × 13 mm.

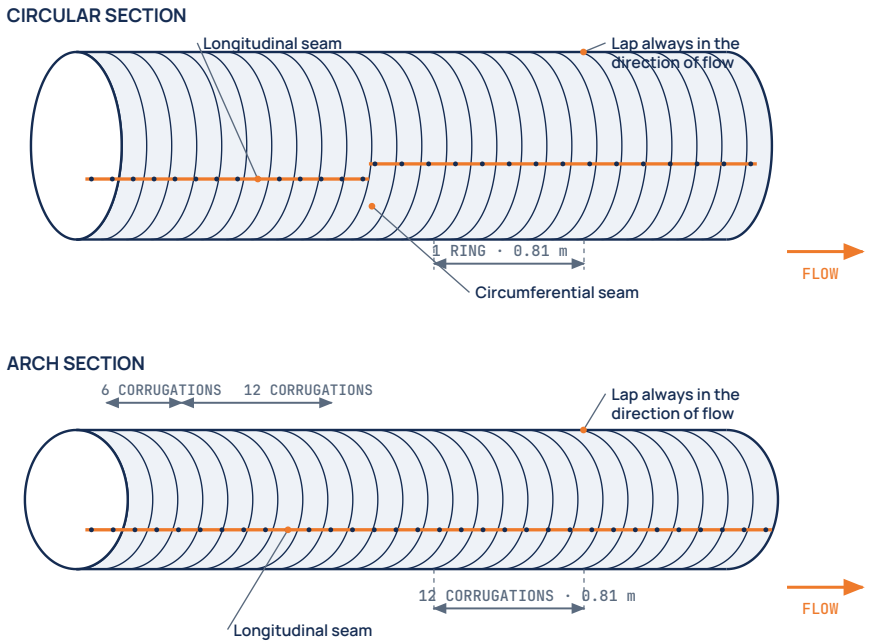
04 — MINIMULTIPLATE SP MP-68

Structures

HS20-44 or HL-93 load combination

The following tables show the main geometric, structural and hydraulic properties of TMC culverts in commercial diameters (span) and, for arch culverts, span and rise, assuming the HS20-44 or HL-93 load combination.

Model	**MiniMultiplate SP MP-68** Corrugation 68 × 13 mm
Shape	
Uses	General use as surface-water drainage. Up to 1.80 m span or diameter.
Thicknesses	e = 1.8 to 3.5 mm
Effective length	L = 0.81 m



SECTIONS Longitudinal and circumferential seams; the lap always runs in the direction of flow.

05 — MINIMULTIPLATE SP MP-68

Properties by section

TABLE 4 · CIRCULAR SECTION

Span (m)	Area (m²)	Thickness (mm)	Weight (kg/m)	Fill height		Flow at 80 % full (m³/s)	V (m/s)
				Min. (m)	Max. (m)		
0.6	0.28	1.8	36.98	0.3	26	0.54	2.14
0.9	0.64	2.0	56.14	0.3	19	1.59	2.80
1.2	1.13	2.5	90.96	0.3	18	2.93	3.00
1.5	1.77	3.0	133.6	0.3	17.5	4.75	3.00
1.8	2.54	3.3	175	0.3	16	6.34	2.82

TABLE 5 · ARCH SECTION

Span (m)	Rise (m)	Area (m²)	Thickness (mm)	Weight (kg/m)	Fill height		Flow at 80 % full (m³/s)	V (m/s)
					Min. (m)	Max. (m)		
0.6	0.41	0.20	1.8	35.8	0.3	5.0	0.29	1.64
0.9	0.56	0.41	2.0	52.1	0.3	4.5	0.73	2.05
1.2	0.79	0.81	2.5	80	0.3	3.5	1.54	2.22
1.5	0.91	1.06	3.0	118	0.3	4.0	2.31	2.45
1.8	1.12	1.64	3.3	155	0.3	4.0	4.00	2.81

Note: fill is measured from the crown of the culvert to the bottom of the subgrade. It can be increased by using thicker plate. Fill material must comply with ASTM A807 and section 502 of MTC EG-2013.

06 — MINIMULTIPLATE SP MP-68

Bolts and nuts

The bolts and nuts of the TMC MiniMultiplate SP MP-68 culvert meet ASTM A449 (bolts) and ASTM A563 Grade C (nuts), and are hot-dip galvanized per ASTM A153.

TABLE 6 · BOLT AND NUT DIMENSIONS

Type	Diameter (in)	Length (in)	Plate thickness (mm)
1	1/2"	7/8"	1.8 to 3.5
2*	1/2"	1 1/4"	

* Only for 3-plate joints.



PLANT Culvert rings ready for shipping.



SITE On-site assembly inspection.

07 — MULTIPLATE SP MP-152 / MP-152 S

Composition and properties

Material requirements per ASTM A929

Chemical composition

The chemical composition requirements of the Multiplate SP MP-152 culvert follow ASTM A929.

TABLE 7 · CHEMICAL COMPOSITION BY HEAT ANALYSIS

Element	Composition, %
Sulfur, max.	0.05
Sum of carbon, manganese, phosphorus, sulfur and silicon, max.	0.75

Mechanical properties

Before corrugation, the base material has the following mechanical properties:

TABLE 8 · MECHANICAL PROPERTIES

Yield strength, min. (MPa)	Tensile strength, min. (MPa)	Elongation in 50 mm, % min.
190	290	25

Dimensions

Nominal plate thicknesses are 2.5 to 8.0 mm (not including the zinc coating). For a specific thickness, please ask our technical team.

TABLE 9 · NOMINAL PLATE DIMENSIONS

Width	Total width (mm)	Net width (mm)	Net length (mm)			No. spaces	No. holes
			4'	6'	8'		
9 FT	854	731	1219.2	1828.8	2438.4	3 N	4
15 FT	1339	1219	1219.2	1828.8	2438.4	5 N	6
18 FT	1587	1463	1219.2	1828.8	2438.4	6 N	7
21 FT	1825	1706	1219.2	1828.8	2438.4	7 N	8

N = 0.2438 m

08 — MULTIPLATE SP MP-152 / MP-152 S

Corrugation dimensions

Nominal corrugation 152 × 51 mm

The plate corrugations form continuous, tangent curves. A corrugation is defined by its pitch (crest-to-crest distance) and depth.

For Multiplate SP MP-152 and SP MP-152 S structures, the nominal corrugation size is 152 × 51 mm. Figure 2 shows its dimensions and tolerances.

NET LENGTHS	NET WIDTHS
1.22 · 1.83 · 2.44 m	**3N · 5N · 6N · 7N**
HOLE SPACING	CORRUGATION
N = 0.2438 m	**152 × 51 mm**

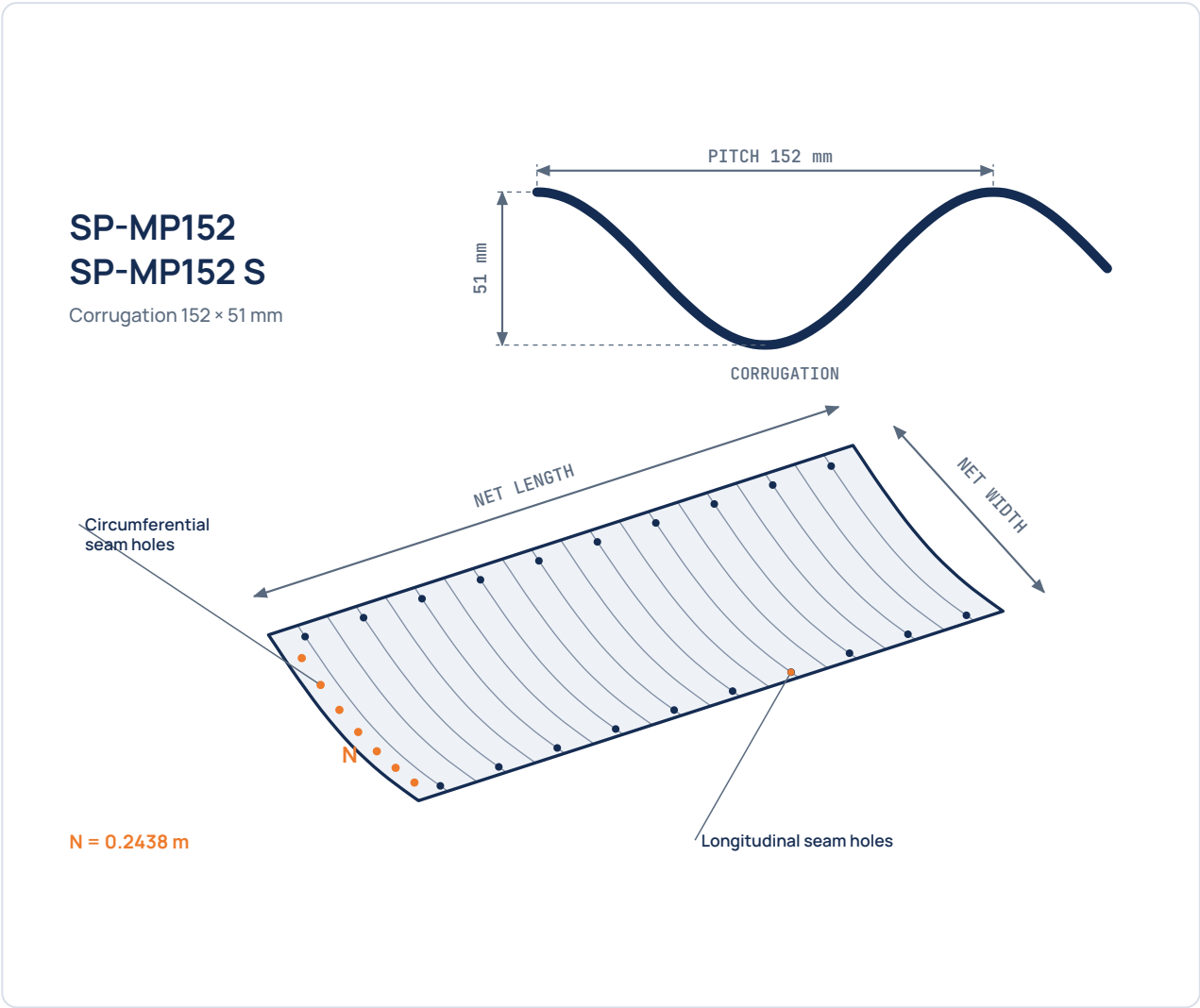


FIGURE 2 Corrugation dimensions 152 × 51 mm.

09 — MULTIPLATE SP MP-152 / MP-152 S

Base channels

Arch channels

The nominal plate thickness for base channels is 6.00 mm. Channels are supplied according to the project design and dimensions; for example, Figure 3 shows an 8-foot (2286 mm) base channel.

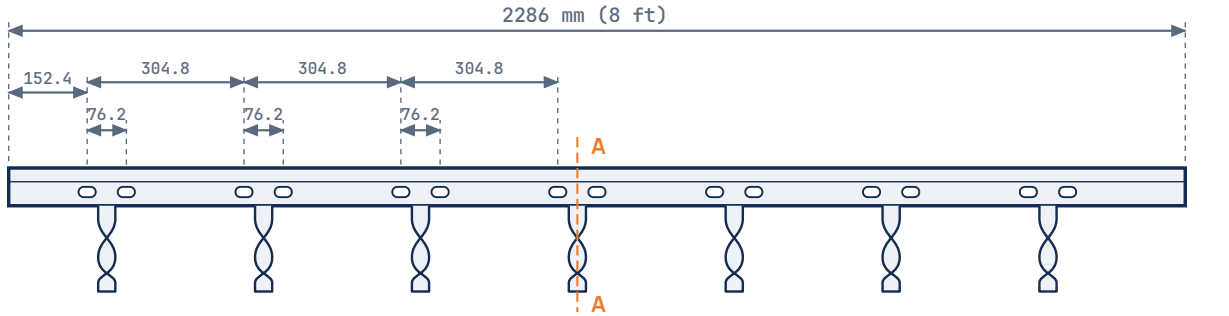


FIGURE 3 Base channel, 2286 mm long (dimensions in mm).

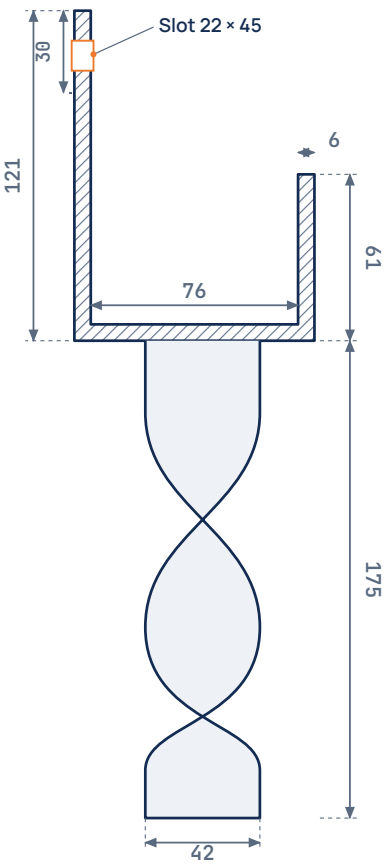


FIGURE 4 Cross section of the base channel (mm).

6.00 mm
Nominal plate thickness

2286 mm
Example length (8 ft)

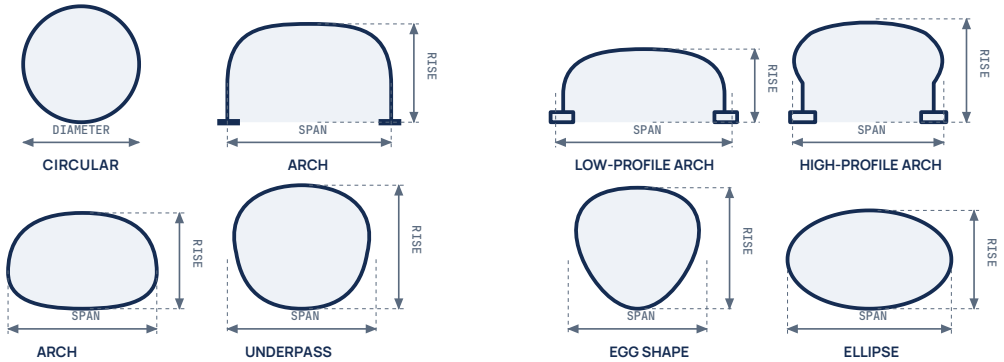
304.8 mm
Anchor spacing

10 — MULTIPLATE SP MP-152 / MP-152 S

Structures

TABLE 10 · BOLTS BY THICKNESS

Diameter	Length (A)	Plate thickness
3/4"	1 1/4"	2.5 to 3.3 mm
3/4"	1 1/2"	2.5 to 8.0 mm
3/4"	1 3/4"	3.3 to 8.0 mm
3/4"	3"	Base channel
3/4"	6"	Headwall connection

Model	Multiplate SP MP-152 152 × 51 mm	Multiplate SP MP-152 S 152 × 51 mm
Shape		
Uses	Ideal for draining small streams and for pedestrian underpasses. Special-case spans up to 8.0 m.	For carrying large flows and for vehicle and rail underpasses. Special-case spans up to 16 m.
Thicknesses	e = 2.5 to 8.0 mm	e = 2.5 to 8.0 mm
Effective length	L = 4', 6' and 8' Net width: 3N, 5N, 6N and 7N	L = 4', 6' and 8' Net width: 3N, 5N, 6N and 7N

11 — MULTIPLATE SP MP-152 / MP-152 S

Bolts and nuts

ASTM A761 standard

The bolts and nuts that join Multiplate SP plates comply with ASTM A761, which specifies:

TABLE 11 · BOLT AND NUT REQUIREMENTS

Feature	Bolts	Nuts
General dimensions	ANSI B18.2.1 Heavy Hex	ANSI B18.2.2 Heavy Hex
Longitudinal and circumferential seams	ASTM A449	ASTM A563 Grade C
Anchoring structures to concrete	ASTM A307	ASTM A563 Grade A
Zinc coating	ASTM A153	ASTM A153
Nominal diameter	3/4"	3/4"



SITE Culvert with concrete headwall.



SITE Erection of long-span culverts.

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Technical advice, quotes and custom fabrication for your next project.

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